UniMind ■ - Phase 1 Technical Audit Report

Architecture, Code Quality, Test Coverage & Security Inspection

Scope: Phase 1 Scaffold & Environment | **Audit Date:** November 1, 2026
Overall Audit Verdict: **PASS (READY FOR PHASE 2)**

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1. Inspection Summary & Verification Checklist

Audit Item	Target Artifact	Findings & Verdict	Status
Dependencies & Versions	requirements.txt	Valid versions; no excess packages. PyMuPDF, FAISS, Sentence-Transformers, Gemini present.	**PASS**
Configuration & Security	.env.example	Zero API keys exposed; full 12-factor settings defined for ports, CORS, and paths.	**PASS**
Application Skeleton	backend/app/main.py	FastAPI factory, modern lifespan, CORS, and versioned routing correctly configured.	**PASS**
Settings Architecture	backend/app/core/config.py	Pydantic BaseSettings with JSON/CSV parser for origins and default fallbacks.	**PASS**
Router Structure	backend/app/routes/api.py	Cleanly modular; isolates v1 endpoints under /api/v1 prefix.	**PASS**
Architectural Integrity	documentation/ARCHITECTURE.md	Fully aligned with code; documents hallucination control and RAG roadmap.	**PASS**
Test Coverage & Rigor	tests/test_health.py	100% of Phase 1 endpoints covered (/, /health, /api/v1/health). All 3 pass.	**PASS**
Secrets / Leaks Check	Entire Repository	Regex scan confirmed 0 exposed API keys, credentials, or private tokens.	**PASS**
Scope Boundaries	RAG / LLM / UI	No premature RAG logic, Gemini client initialization, or React frontend created.	**PASS**
GitHub Integrity	Git status / diff	Original README.md, LICENSE, and .gitignore remain 100% intact.	**PASS**

2. Identified Issues & Severity Analysis

ID	Component	Observation / Potential Risk	Severity	Recommendation
ISS-01	requirements.txt	python-multipart is omitted. Required by FastAPI when handling multipart form uploads (e.g. PDF document uploads in Phase 2).	**LOW** (Phase 2)	Add python-multipart>=0.0.9 before implementing PDF upload endpoint.
ISS-02	config.py / main.py	ALLOWED_ORIGINS validator falls back to ['*'] on invalid formats. When allow_credentials=True, browsers reject wildcard origin.	**LOW** (Security)	Fallback to explicit defaults ['http://localhost:3000'] instead of wildcard ['*'].
ISS-03	config.py	GEMINI_API_KEY is typed as plain str rather than SecretStr. Potential accidental leakage in debug logs or repr().	**INFO** (Best Practice)	Upgrade to pydantic.SecretStr in Phase 3 when LLM authentication is wired.
ISS-04	main.py vs health.py	Duplicate health response logic: /health defined in main.py and /api/v1/health defined in health.py.	**INFO** (Code Quality)	Refactor main.py to reuse health_check() or mount root health router directly.

3. Test Suite Re-Execution Results

```
Test Command: python -m unittest discover -s tests -v
Test Execution Summary:
• test_api_v1_health_check (test_health.TestHealthEndpoints) ... ok
• test_root_endpoint (test_health.TestHealthEndpoints) ... ok
• test_root_health_check (test_health.TestHealthEndpoints) ... ok
Status: 3 Passed, 0 Failed, 0 Errors in 0.014s  |  Conclusion: 100% Endpoint Health Verified.
```